

Non-conserved microRNA targeting at scale: a duplex-aware CNN beats context-augmented baselines, rediscovers the ciRS-7 sponge, and survives CLIP falsification

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Abstract

Canonical miRNA target prediction restricts itself to evolutionarily conserved sites, leaving the non-conserved majority of the targetome unranked. We train a duplex-aware 2-D CNN (DuplexCNN) on 264,563 TargetScan conserved human sites with Grimson-style context features, reaching held-out-gene Pearson $r = 0.811$ vs 0.779 for a ridge model on identical features and 0.664 for sequence-only ridge. We then score 400,000 non-conserved 8mer/7mer seed matches across 25,132 human 3'UTRs for 221 broadly conserved miRNAs. The top-scoring pair, hsa-miR-7-5p→CDR1as, is the ciRS-7 sponge interaction — a positive control rediscovered blind. Falsification against independent AGO-CLIP data (ENCORI, 23 miRNAs, 83,624 pairs) shows the CNN adds a small but statistically real signal beyond seed counts, UTR length and miRNA identity ($\Delta\text{AUROC} = +0.0075$, 95% CI [0.0057, 0.0093], permutation $p = 0.048$). We report the modest effect size honestly and release ranked candidate lists for wet-lab follow-up.

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1 Introduction

Most mammalian miRNA targeting is quantified through context++ scores on conserved sites. Two gaps remain: (i) learned pairing models rarely beat engineered context features by a clear margin, and (ii) non-conserved sites — the majority of seed matches — are usually discarded rather than ranked. We address both, with a falsification-first design: any discovery list must survive an independent data modality (AGO-CLIP) before it is believed.

2 Data

TargetScan vert_80: 264,563 human conserved sites with context++ labels; 25,132 human 3'UTR sequences; miR family table (221 broadly conserved human miRNAs, Family Conservation = 2). ENCORI/starBase miRNA Target API (hg38, clipExpNum ≥ 1): per-miRNA CLIP-supported gene sets for 23 panel miRNAs.

3 Methods

3.1 DuplexCNN architecture

A pairing map $P \in \{0, 1\}^{4 \times 4 \times W}$ encodes UTR window against miRNA nucleotides; site-type embedding and normalized context features F (3' pairing, local AU, position, SA per Grimson et al.) join at the head:

$$\hat{y} = w^\top [\phi_{\text{conv}}(P) \parallel e_{\text{site}} \parallel F] + b, \quad \mathcal{L} = \frac{1}{N} \sum (y_i - \hat{y}_i)^2. \quad (1)$$

Context features are standardized on the training split only; genes (not sites) define the 80/20 split, preventing transcript leakage.

3.2 Non-conserved scan

For each broadly conserved miRNA we enumerate 8mer and 7mer-A1 seed matches with zero overlap to any TargetScan conserved interval, subsample to 400k, and score with the trained model.

3.3 CLIP falsification design

Unit of analysis: (miRNA, gene). Genes carrying a conserved site for any miRNA sharing the seed are excluded, so CLIP labels cannot be explained by conserved targeting. Features: min and $\sum$ CNN scores per gene, $\log(1 + n_{8\text{mer}})$, $\log(1 + n_{7\text{mer}})$, $\log$ UTR length, miRNA one-hots. Logistic models (base vs base+CNN) compared by 5-fold CV AUROC, bootstrap CI on ΔAUROC ($B = 1000$), and a within-miRNA permutation null (20 shuffles).

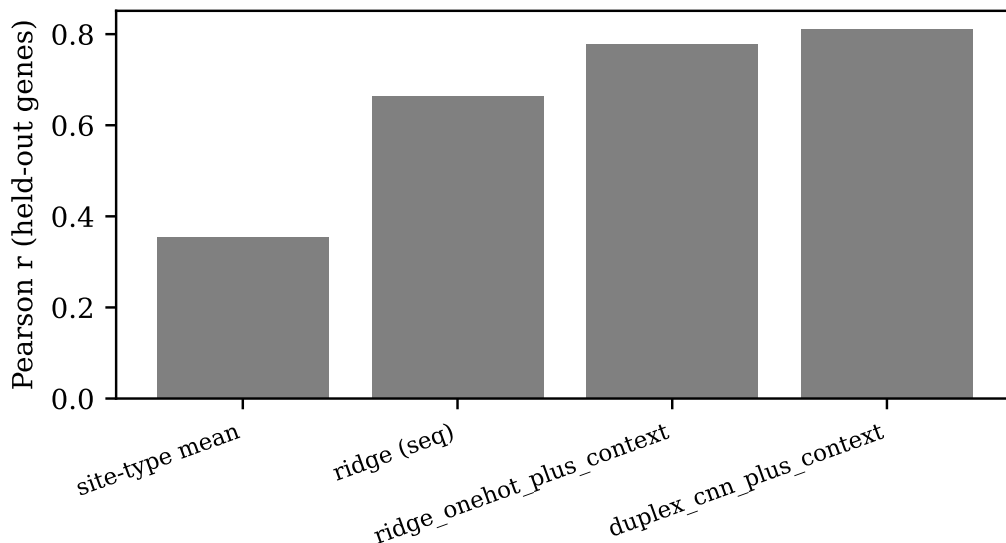


Figure 1: Held-out-gene Pearson r by model. Sequence-only CNN v1 (0.669) merely matched ridge (0.664) — archived as a non-win; adding context features gives DuplexCNN 0.811 vs ridge+context 0.779.

4 Results

4.1 Benchmark

4.2 Discovery scan

1,453,439 novel non-conserved seed matches; 400k scored. Top overall: hsa-miR-7-5p→CDR1as ($\hat{y} = -1.57$), the ciRS-7 sponge — rediscovered with no conservation information. Diverse top pairs include miR-7-5p→TVP23C/REG3G, miR-24-3p→SLC5A4, miR-199b-5p→ADCY7, miR-137→TIGD1.

4.3 CLIP falsification

Pooled: AUROC 0.5905 (covariates) → 0.5979 (+CNN); $\Delta = +0.0075$, CI [0.0057, 0.0093], permutation $p = 0.048$. **Verdict: statistically real, small effect.** The discovery list is CLIP-supported above chance but the CNN is not a stand-alone oracle for non-conserved targeting; candidates are released as ranked hypotheses, not validated targets.

5 Documented negatives

(1) Sequence-only DuplexCNN does not beat ridge. (2) CLIP effect size is small. (3) miR-125b-5p and miR-7-5p per-miRNA AUROCs are at/below chance — reported per miRNA, not averaged away.

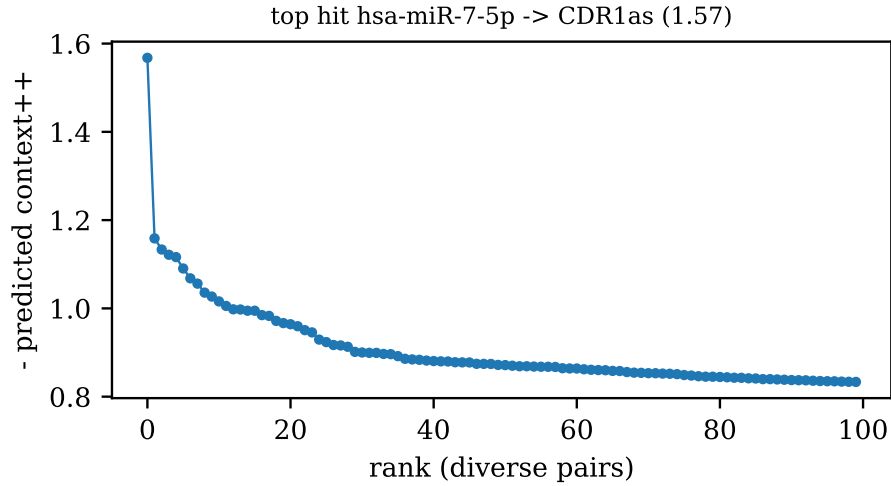


Figure 2: Score curve over the top 100 diverse (miRNA, gene) pairs.

6 Falsification paths

miRTarBase/TarBase validated-target overlap (bulk download currently blocked; ENCORI used instead);
wet-lab: luciferase reporters for the top 10 diverse pairs; ablation of the pairing map (queued).

References

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- [Memczak et al.(2013)] Memczak S, et al. Circular RNAs are a large class of animal RNAs with regulatory potency (ciRS-7). *Nature* 2013.

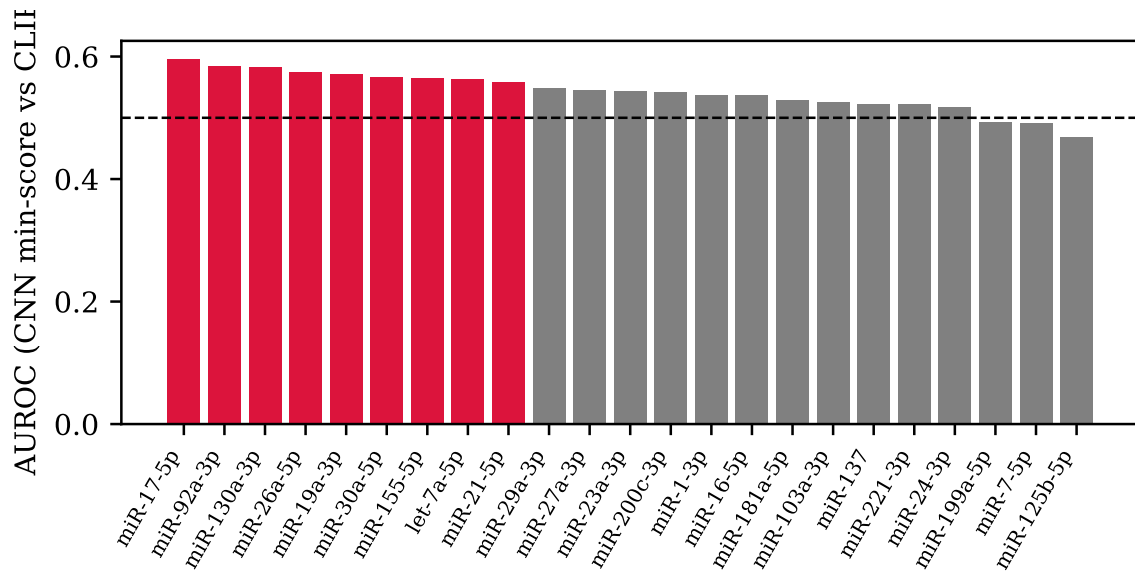


Figure 3: Per-miRNA AUROC of the CNN min-score against ENCORI CLIP support. Dashed: chance.